

A Study of User Interaction with Large Language Models in Home AI-ECG Device

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Abstract—Background: The widespread adoption of portable electrocardiogram (ECG) devices has created a significant gap in the interpretation of complex medical data between users and professionals. Home-based AI ECG devices equipped with Large Language Models (LLMs) have emerged as a potential tool to bridge this gap. This study aims to explore user behavior after receiving portable ECG results, focusing on their interaction with LLMs, including conversation intent, retention, and engagement.

Methods: This study conducted a retrospective analysis of 480,532 real-world dialogue records from the “Wenxin Wuyang” home AI ECG device, collected between April 16, 2024, and August 18, 2025. After cleaning and preprocessing, 109,017 independent dialogue sessions were integrated. A random sample of 1% of these sessions (1,091 groups) was selected for analysis, using a “Large Model-assisted + manual review” approach to examine user intent. Additionally, we conducted statistical analysis on user retention rates and the number of interaction turns after receiving AI interpretation results, and longitudinally tracked the interaction depth evolution of retained users.

Results: Intent analysis revealed that the three most frequent intents were “Result interpretation” (81.30%), “Treatment suggestions” (51.15%), and “Inquiries about causes” (41.89%). Over 87% of dialogues contained two or more dimensions of intent. Retention analysis showed that 54.21% of users engaged in multiple conversations. Regarding interaction depth, most dialogues (63.5%) were moderately interactive, with 2-5 turns, while shallow inquiries with just one turn accounted for 24.38%. Longitudinal analysis of retained users revealed a J-shaped evolution in interaction depth. The first conversation had the deepest interactions (average of 3.51 turns), which then decreased and stabilized at a utility baseline (approximately 3.05 turns). Subsequently, the interaction depth gradually increased with more frequent use.

Conclusion: The study confirms that Large Language Models are playing a central role in helping users bridge the gap in interpreting portable ECG data. Users exhibit complex interaction patterns centered around result interpretation, with multi-intent behavior and a willingness to engage in long-term usage. These findings provide empirical evidence for optimizing digital health services and highlight the potential risks associated with the frequent demand for treatment suggestions.

Index Terms—Large Language Model, Portable Electrocardiogram, Interpretation Gap, User Behavior, Real-world Study

I. INTRODUCTION

The widespread adoption of portable and wearable health monitoring devices, particularly the rise of portable electrocardiogram (ECG) devices, has significantly enhanced the ability of ordinary consumers to collect physiological data that was previously confined to clinical settings. This shift has allowed users to transition from being passive receivers of medical information to becoming active participants in managing their own health. Empowering individuals with such health-monitoring capabilities is one of the core goals of the digital health revolution [1] [2].

However, when faced with complex medical data, many users lack the necessary medical training to understand the clinical significance of these results [3]. This creates a substantial gap between data acquisition and data interpretation, known as the interpretation gap [4]. Although existing portable ECG devices often connect to cloud platforms and AI-based systems to automatically present results [5] [6], users still struggle to comprehend abnormal results or seek further clarification on underlying data and medical terms, particularly when they need additional support or professional advice [4].

In this context, Large Language Models (LLMs) have emerged as a potential solution. Conversations between users and LLMs have become a key pathway for bridging the interpretation gap, even demonstrating empathy in their responses to users [7]. On the one hand, this behavior pattern presents an unprecedented opportunity to improve public health literacy and achieve personalized health management [8]. On the other hand, it also introduces new challenges. Both the industry and academia are focused on the accuracy, safety, and inherent limitations of LLM responses, particularly the potential risks of non-medical advice [9] [10]. Therefore, before deeply assessing these specific risks, it is essential to first analyze the

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actual interaction patterns and core demands of users when using such tools.

However, large-scale empirical research on user behavior and interaction in specific dialogues between users and LLMs after using portable devices remains scarce. Existing studies tend to focus more on the use of LLMs in professional clinical settings or explore their various tasks in the medical field from a macro perspective [11] [12]. Thus, this study focuses on the user behavior exhibited by ordinary users after receiving ECG data from portable ECG devices and AI interpretation results, specifically analyzing user intentions, dialogue frequency, and interaction patterns in conversations with LLMs.

II. METHODS

A. Data Source and Preprocessing

This project is a retrospective study based on real-world dialogue data. The dataset was sourced from the “Wenxin Wuyang” home AI ECG device database, which includes 480,532 dialogue records generated by users in China interacting with the “Wenxin Wuyang” AI model between April 16, 2024, and August 18, 2025. To ensure data quality and the effectiveness of the analysis, we performed data preprocessing. First, we removed all dialogue records initiated solely by the system, where no response was provided by the user. Additionally, we employed rule-based and keyword matching methods to filter out clearly meaningless test and invalid dialogues (e.g., entries containing only “Test” or a single punctuation mark) and dialogues with incomplete information. Subsequently, to better analyze continuous interaction behaviors, we grouped the data into conversation sessions. Specifically, dialogues from the same user ID within a 30-minute window were defined as a single independent conversation session. If the time interval between two consecutive user statements exceeded 30 minutes, the latter was considered the start of a new session. This research plan has been reviewed and approved by the Peking University Medical Ethics Committee (Approval No.: IRB00001052-23071) and was granted an exemption from requiring signed informed consent.

B. Random Sampling and User Intent Analysis

To delve deeper into the core intentions behind users’ interactions with the AI model, we conducted a 1% random sampling of the preprocessed conversation dataset. This sampling ratio was chosen to ensure sample representativeness while balancing the cost and feasibility of manual review. We adopted a hybrid approach, combining “Large Model Assistance + Manual Review,” to classify user intents. Specifically, we first utilized the GLM-4.5 model from ZhiPu AI to automatically categorize all user questions in each conversation session. Then, researchers manually reviewed the model’s categorization results to ensure the final classification’s accuracy and reliability. After completing the intent classification for all sampled data, we used descriptive statistical methods to analyze the frequency distribution of different user intents.

C. User Retention and Engagement Analysis

To assess the long-term use of the AI model by users and the depth of single-session interactions, we analyzed both user retention and user engagement. User retention was defined as the proportion of users who initiated more than one independent conversation session (≥ 2) during the study period. This was calculated by counting the number of conversation sessions associated with each unique user ID. User engagement was measured by the number of interaction turns in each individual conversation session. One interaction turn was defined as one user question followed by one model response. Based on the number of interaction turns, we categorized user engagement into four levels: 1 turn: Shallow inquiry, usually simple information gathering; 2-5 turns: Moderate interaction, with users asking follow-up questions or seeking clarification; 6-9 turns: Deep interaction, with users showing sustained exploration interest; 10 or more turns: High engagement, with extensive or in-depth discussion of the topic.

D. Longitudinal Evolution Analysis of Retained Users

To explore the behavioral evolution patterns of retained users, we conducted a longitudinal analysis of how the number of interaction turns changed in subsequent conversation sessions. First, we selected all conversation data from users who had multiple dialogues. We then calculated the average interaction turns across different session orders for each user, in order to observe how the interaction depth evolved over time.

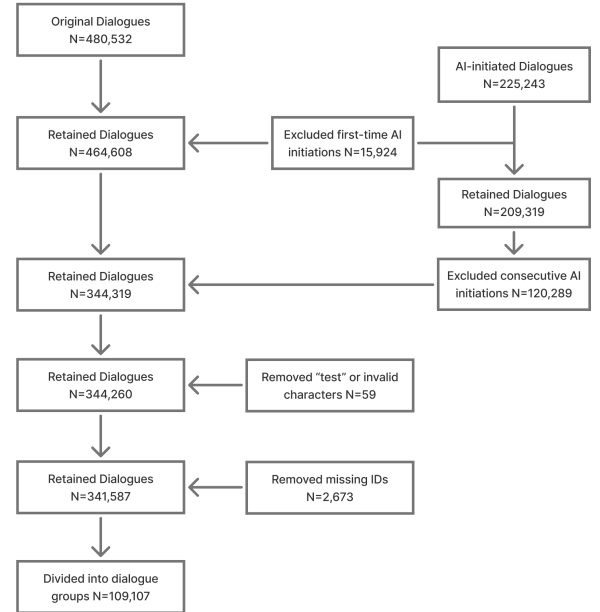


Fig. 1. Data Cleaning and Grouping Process

III. RESULTS

A. Data Cleaning and Grouping

The results of data cleaning and grouping are shown in Figure 1. Among the initial 480,532 dialogue records, 15,924 were single-turn system-initiated queries, and 120,289 were system-initiated multi-turn queries. In addition, 59 invalid dialogues and 2,673 incomplete dialogues were identified and removed during the cleaning process.

After these steps, a total of 341,587 valid dialogue records were retained. Based on this dataset, dialogues from the same user within a 30-minute time window were grouped as a single session, resulting in 109,017 organized conversation sessions.

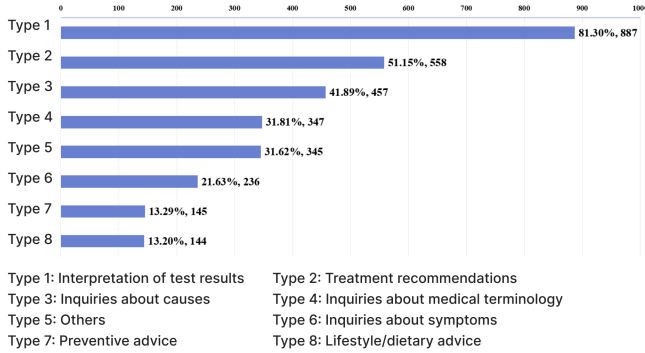


Fig. 2. Statistical analysis of sampling question types

B. User Intent Analysis

After a 1% random sampling of the dialogue groups, a total of 1,091 dialogue groups were obtained. Based on the classification results of the GLM-4.5 model, all user queries were categorized into eight types: Interpretation of test results, Inquiries about medical terminology, Treatment recommendations, Inquiries about causes, Preventive advice, Inquiries about symptoms, Lifestyle/dietary advice, and Others. After manual verification, the detailed distribution is shown in Figure 2. Among them, “Interpretation of test results” accounted for the largest proportion with 887 groups (81.30%), followed by “Treatment recommendations” with 558 groups (51.15%), “Inquiries about causes” with 457 groups (41.89%), “Inquiries about medical terminology” with 347 groups (31.81%), “Others” (including irrelevant or unclassifiable content) with 345 groups (31.62%), “Inquiries about symptoms” with 236 groups (21.63%), “Preventive advice” with 145 groups (13.29%), and “Lifestyle/dietary advice” with 144 groups (13.20%).

As for the distribution of dialogue group types (Table 1) reveals that users typically explore multiple dimensions of questions within a single conversation. The statistical results show that: dialogues containing only 1 question type accounted for 236 groups, representing 21.63%; dialogues with 2 question types comprised 325 groups, or 29.79%; dialogues containing 3 question types made up 283 groups, representing 25.94%; 4 question-type dialogues totaled 162

groups, accounting for 14.85%; and dialogues with 5 or more question types comprised 85 groups, or 7.79%.

TABLE I
QUESTION TYPE DIVERSITY AND DISTRIBUTION

Question-Type Diversity	Count (Groups)	Percentage (%)
1 type of question	236	21.63
2 types of questions	325	29.79
3 types of questions	283	25.94
4 types of questions	162	14.85
5 or more types of questions	85	7.79

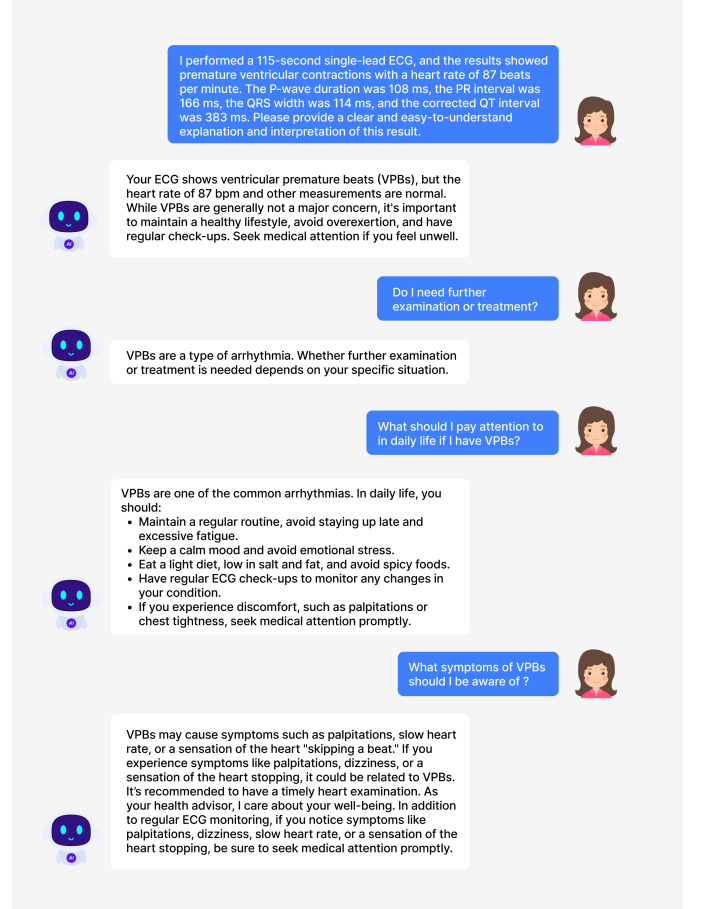


Fig. 3. User-AI Large Language Model Dialogue Case

C. User Retention and Interaction Turns

According to the user ID statistics, a total of 26,419 users were identified. Overall, 12,097 users engaged in only a single dialogue, accounting for 45.79% of all users, while 14,322 users participated in multiple dialogues, representing 54.21%. On average, each user interacted with the model 4.13 times (SD = 8.19; median = 2), with the maximum reaching 580 dialogues, indicating the presence of significant outliers.

In terms of interaction turns, each dialogue group contained an average of 3.13 turns (SD = 2.81; median = 2), with the maximum reaching 108 turns. Among them, 26,578 groups

contained only a single turn, accounting for 24.38% of all dialogues, suggesting a considerable proportion of shallow interactions. The majority (69,230 groups, 63.50%) fell within 2–5 turns, reflecting moderate exchanges between users and the model. Dialogue groups with 6–9 turns numbered 9,838 (9.03%), representing a certain degree of in-depth discussion, while those with 10 or more turns were 3,371 (3.09%), indicating a small fraction of highly engaged interactions.

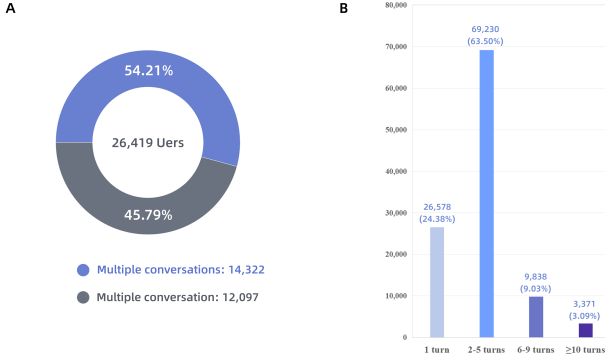


Fig. 4. A: User retention rate statistics; B: Conversation turn count statistics

D. Longitudinal Evolution of Interaction Depth

This study analyzed the longitudinal evolution of interaction depth among 14,322 retained users, as shown in Figure 5. The results indicate that the average interaction depth did not follow a linear pattern but instead exhibited a J-shaped trajectory. During the first session, users showed the highest level of interaction, with an average of 3.51 turns. The depth then declined rapidly to 3.07 turns in the second session and stabilized around 3.05 turns by the fourth session, forming a utility baseline.

After this baseline, interaction depth entered a phase of slow yet steady growth, gradually increasing from 3.07 turns in the fifth session to 3.33 turns by the nineteenth session. However, a noticeable drop in interaction depth was observed between the fourteenth and sixteenth sessions.

In addition, the number of active users decreased progressively with each session. Among the initial 14,322 users, only 5,793 participated in the fifth session, 2,486 in the tenth session, 1,339 in the fifteenth session, and merely 839 continued to the twentieth session. a

IV. DISCUSSION

This study is the first to systematically analyze user behaviors—including intentions and engagement patterns—after receiving AI-interpreted results from portable electrocardiogram (ECG) devices, using large-scale real-world dialogue data. The findings clearly demonstrate that large language models (LLMs) are playing a crucial role in bridging the interpretation gap between ordinary users and complex medical data.

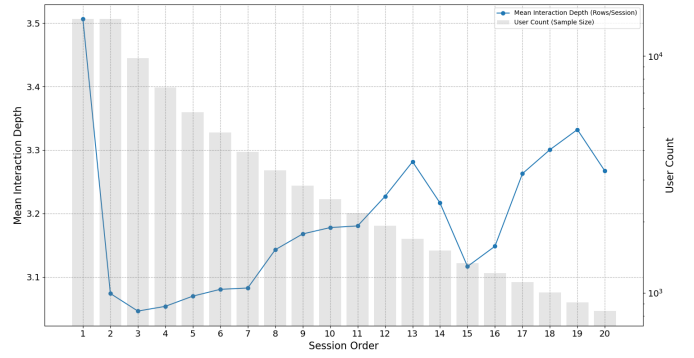


Fig. 5. Evolution of interaction depth among retained users. The line plot illustrates the change in interaction depth (number of turns) with increasing session count, while the gray bars represent the corresponding variation in user numbers across sessions.

A. Findings and Interpretation

A substantial 81.30% of user intents focused on “Interpretation of test results”, directly confirming that users have an urgent need for professional interpretive support after obtaining physiological data. This highlights that AI dialogue systems have become one of the preferred tools for users seeking immediate and personalized health information. Meanwhile, “Treatment recommendations” (51.15%) and “Inquiries about causes” (41.89%) followed closely, suggesting that users’ needs extend beyond mere result interpretation toward deeper understanding and actionable guidance.

An important observation is that over 78% of dialogues contained two or more types of intents, indicating that users’ consultation processes are multidimensional and non-linear. Users often begin by asking about their test results, then naturally progress to terminology explanations, cause exploration, preventive strategies, and lifestyle adjustments within the same conversation. Such complex interaction patterns place high demands on models’ multi-turn dialogue capacity, contextual understanding, and knowledge integration—underscoring the potential of LLMs as an integrated entry point for digital health consultations.

In terms of user engagement, most dialogues (63.5%) represented moderate interactions (2–5 turns), while 24.38% were shallow inquiries, showing that many users still use the model primarily for quick answers. This moderate interaction–dominant pattern, combined with the finding that 54.21% of users engaged in multiple sessions, suggests that AI models are not merely one-time query tools but have successfully served as long-term health companions for a substantial subset of users.

To further examine the behavioral dynamics of these 54.21% retained users, the longitudinal analysis revealed a clear J-shaped trend. Specifically, users exhibited the highest exploratory engagement in the first session (mean = 3.51 turns), likely reflecting their initial exploration of the model’s capabilities. The interaction depth then rapidly stabilized around 3.05 turns by the fourth session, forming a utility baseline. This shift indicates a behavioral transition from initial exploration

to instrumental use of the AI model. Most notably, after this baseline, interaction depth entered a slow but steady growth phase, reaching 3.33 turns by the nineteenth session—an increase of 9.18%—suggesting that high-retention users gradually deepened their engagement over time.

B. Implications and Practical Significance

The findings offer valuable insights for the design and optimization of digital health services. First, they underscore the central importance of result interpretation as a core function. Developers should continuously enhance model capabilities in understanding and explaining ECG reports, key metrics, and abnormal findings. Second, given the diversity and interrelated nature of user intents, models should be designed to proactively guide and extend dialogues—for example, after explaining a test result, the system could offer relevant information about causes, preventive measures, or lifestyle recommendations—thus creating a more intelligent and comprehensive service loop.

Additionally, the high frequency of treatment-related inquiries serves as an important cautionary signal. While LLMs can technically generate plausible advice, such outputs may pose significant safety risks [13]. This highlights the need for platforms to integrate clear risk disclaimers and non-medical advisory statements, and to explore safe and compliant pathways that guide users toward professional healthcare services.

C. Limitations and Future Directions

Several limitations should be acknowledged. First, this is a retrospective study that analyzed user query behaviors but did not evaluate the accuracy, safety, or user satisfaction with the model's responses. Second, demographic and clinical background information (e.g., age, gender, education, medical history) was unavailable, although these factors could strongly influence user behavior. Moreover, intent analysis was conducted on a 1% random sample, which ensured feasibility but may introduce sampling bias.

The longitudinal analysis also reflects the natural attrition of user cohorts: as the number of sessions increased, the number of retained users declined from 14,322 in the first session to 839 by the twentieth. Therefore, the upward trend observed in the latter part of the J-shaped curve primarily reflects the behavior of highly engaged users rather than the entire cohort. In addition, the smaller sample size in later sessions may lead to normal statistical fluctuations (at sessions 14–16), and thus these findings should be interpreted as representing overall trends rather than precise session-level effects.

Future research could proceed in several directions. First, systematic evaluation of model response quality—including accuracy, completeness, and safety—is essential to assess its true clinical utility. Second, prospective studies or user surveys could be conducted to investigate users' motivations, satisfaction, and the real-world impact of AI-generated responses on health knowledge and behavior. Third, integrating user profiling data would allow for the identification of behavioral differences among various user subgroups, enabling more

personalized and precise service design. Finally, exploring human–AI collaborative models is an important next step, examining how AI dialogue systems can be effectively integrated into healthcare workflows—as supportive tools for clinicians rather than potential sources of risk.

D. Conclusion

In summary, by analyzing large-scale real-world dialogue data, this study demonstrates that AI-based large language models play a positive and essential role in helping users bridge the interpretation gap in portable ECG use. Users exhibited a complex, multi-intent interaction pattern centered on result interpretation and showed a sustained willingness to engage with the system over time. These findings provide valuable empirical evidence for improving digital health product design and enhancing user health literacy, while also reminding us to remain vigilant about the ethical and safety challenges accompanying such applications.

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